

Часть 1

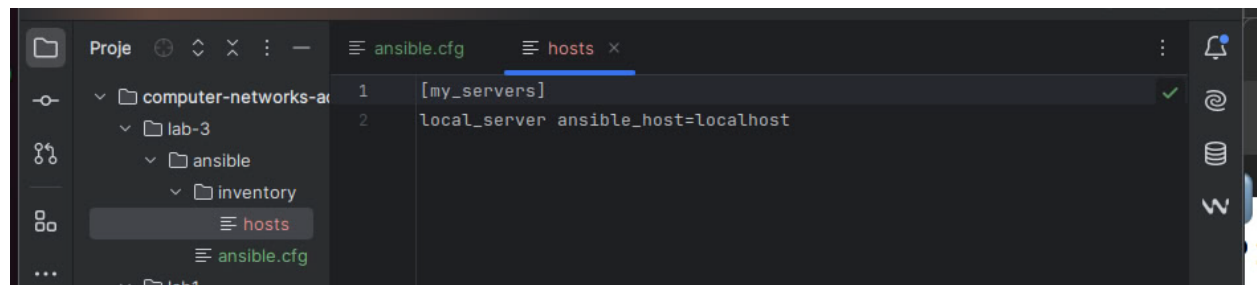
Скачаем pip

```
undercat@undercat-Aspire-A315-59: /GolandProjects/computer-networks-administration$ curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py && python3 get-pip.py
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Done    % Upload   Total       Spent    Left     Speed
100 2131k  100 2131k    0     0 2943k      0  0:00:00  0:00:00  0:00:00 2939k
Defaulting to user installation because normal site-packages is not writeable
Collecting pip
  Downloading pip-25.3-py3-none-any.whl.metadata (4.7 kB)
Collecting setuptools
  Downloading setuptools-80.9.0-py3-none-any.whl.metadata (6.6 kB)
Collecting wheel
  Downloading wheel-0.45.1-py3-none-any.whl.metadata (2.3 kB)
  Downloading pip-25.3-py3-none-any.whl (1.8 MB)
  Downloading setuptools-80.9.0-py3-none-any.whl (1.2 MB)
  Downloading wheel-0.45.1-py3-none-any.whl (72 kB)
Installing collected packages: wheel, setuptools, pip
WARNING: The script wheel is installed in '/home/undercat/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
WARNING: The scripts pip, pip3 and pip3.10 are installed in '/home/undercat/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-25.3 setuptools-80.9.0 wheel-0.45.1
undercat@undercat-Aspire-A315-59: /GolandProjects/computer-networks-administration$ python3 -m pip install ansible
Defaulting to user installation because normal site-packages is not writeable
Collecting ansible
  Downloading ansible-10.7.0-py3-none-any.whl.metadata (8.0 kB)
Collecting ansible-core==2.17.7 (from ansible)
  Downloading ansible_core-2.17.14-py3-none-any.whl.metadata (7.0 kB)
Collecting jinja2>=3.0.0 (from ansible-core==2.17.7->ansible)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: PyYAML>=5.1 in /usr/lib/python3/dist-packages (from ansible-core==2.17.7->ansible) (5.4.1)
Requirement already satisfied: cryptography in /usr/lib/python3/dist-packages (from ansible-core==2.17.7->ansible) (3.4.8)
Collecting packaging (from ansible-core==2.17.7->ansible)
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting resolvelib<1.1.0,>=0.5.3 (from ansible-core==2.17.7->ansible)
  Downloading resolvelib-1.0.1-py2.py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/lib/python3/dist-packages (from jinja2>=3.0.0->ansible-core==2.17.7->ansible) (2.0.1)
Downloading ansible-10.7.0-py3-none-any.whl (51.6 MB)
  Downloading ansible_core-2.17.14-py3-none-any.whl (2.2 MB)
  Downloading resolvelib-1.0.1-py2.py3-none-any.whl (17 kB)
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
  Downloading packaging-25.0-py3-none-any.whl (66 kB)
Installing collected packages: resolvelib, packaging, jinja2, ansible-core, ansible
  3/5 [ansible-core] WARNING: The scripts ansible, ansible-config, ansible-connection, ansible-console, ansible-
  laybook, ansible-pull and ansible-vault are installed in '/home/undercat/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  4/5 [ansible] WARNING: The script ansible-community is installed in '/home/undercat/.local/bin' which is not o
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
```

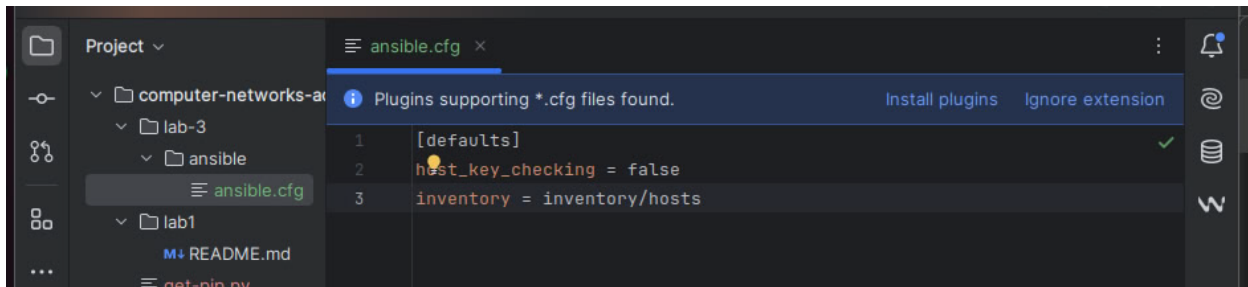
Скачаем ansible

```
undercat@undercat-Aspire-A315-59: /GolandProjects/computer-networks-administration$ python3 -m pip install ansible
Defaulting to user installation because normal site-packages is not writeable
Collecting ansible
  Downloading ansible-10.7.0-py3-none-any.whl.metadata (8.0 kB)
Collecting ansible-core==2.17.7 (from ansible)
  Downloading ansible_core-2.17.14-py3-none-any.whl.metadata (7.0 kB)
Collecting jinja2>=3.0.0 (from ansible-core==2.17.7->ansible)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Requirement already satisfied: PyYAML>=5.1 in /usr/lib/python3/dist-packages (from ansible-core==2.17.7->ansible) (5.4.1)
Requirement already satisfied: cryptography in /usr/lib/python3/dist-packages (from ansible-core==2.17.7->ansible) (3.4.8)
Collecting packaging (from ansible-core==2.17.7->ansible)
  Downloading packaging-25.0-py3-none-any.whl.metadata (3.3 kB)
Collecting resolvelib<1.1.0,>=0.5.3 (from ansible-core==2.17.7->ansible)
  Downloading resolvelib-1.0.1-py2.py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/lib/python3/dist-packages (from jinja2>=3.0.0->ansible-core==2.17.7->ansible) (2.0.1)
Downloading ansible-10.7.0-py3-none-any.whl (51.6 MB)
  Downloading ansible_core-2.17.14-py3-none-any.whl (2.2 MB)
  Downloading resolvelib-1.0.1-py2.py3-none-any.whl (17 kB)
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
  Downloading packaging-25.0-py3-none-any.whl (66 kB)
Installing collected packages: resolvelib, packaging, jinja2, ansible-core, ansible
  3/5 [ansible-core] WARNING: The scripts ansible, ansible-config, ansible-connection, ansible-console, ansible-
  laybook, ansible-pull and ansible-vault are installed in '/home/undercat/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  4/5 [ansible] WARNING: The script ansible-community is installed in '/home/undercat/.local/bin' which is not o
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
```

Заполним хосты — машины, на которых будет работать ansible



Заполним конфиг ansible. Пропишем, что не хотим проверять ssh-ключи хостов, и прокинем директорию машин, на которых будет запущен ansible.



Проверим, жив ли ansible и подключен ли он к клиенту.

```
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ ansible
le my_servers -m ping -c local
[WARNING]: Platform linux on host local_server is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
local_server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
  },
  "changed": false,
  "ping": "pong"
}
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$
```

Выполним команду `ansible my_servers -c local -m shell -a 'echo test_file_content > $HOME/test.txt'`

Проверим, что файл создался в указанной директории.

```
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ cd ..
../../../../
undercat@undercat-Aspire-A315-59:~$ nano test
test-mic.wav test.txt
undercat@undercat-Aspire-A315-59:~$ nano test.txt
undercat@undercat-Aspire-A315-59:~$ cat test.txt
test_file_content
undercat@undercat-Aspire-A315-59:~$
```

Часть 2

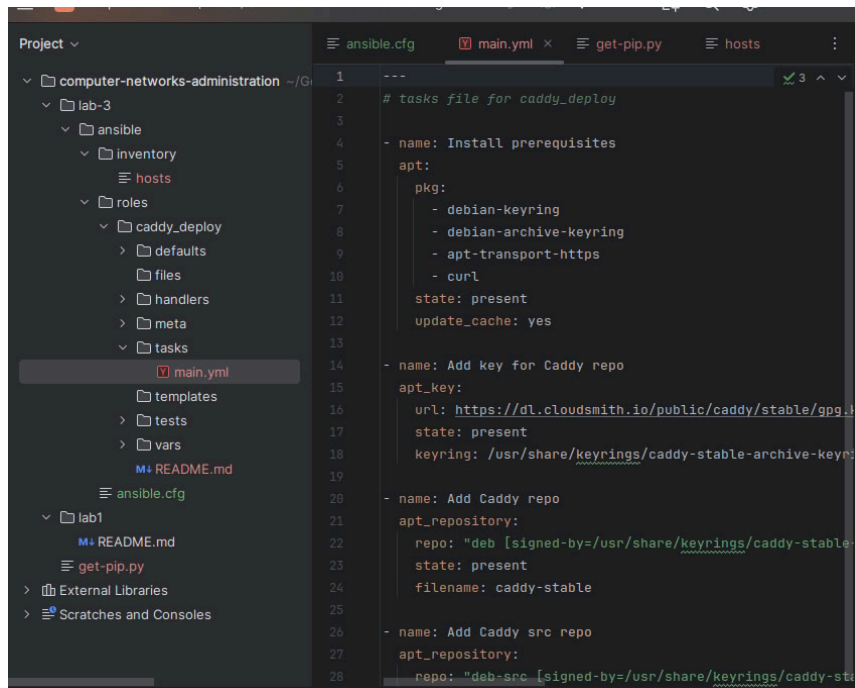
Инициализируем конфигурационное дерево caddy_deploy

```
undercat@undercat-Aspire-A315-59:~$ cd GoLandProjects/computer-networks-administration/lab-3/ansible
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ mkdirroles
mkdirroles: command not found
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ mkdir
roles
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ cd roles/
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles$ ansible-galaxy init caddy_deploy
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles$
```

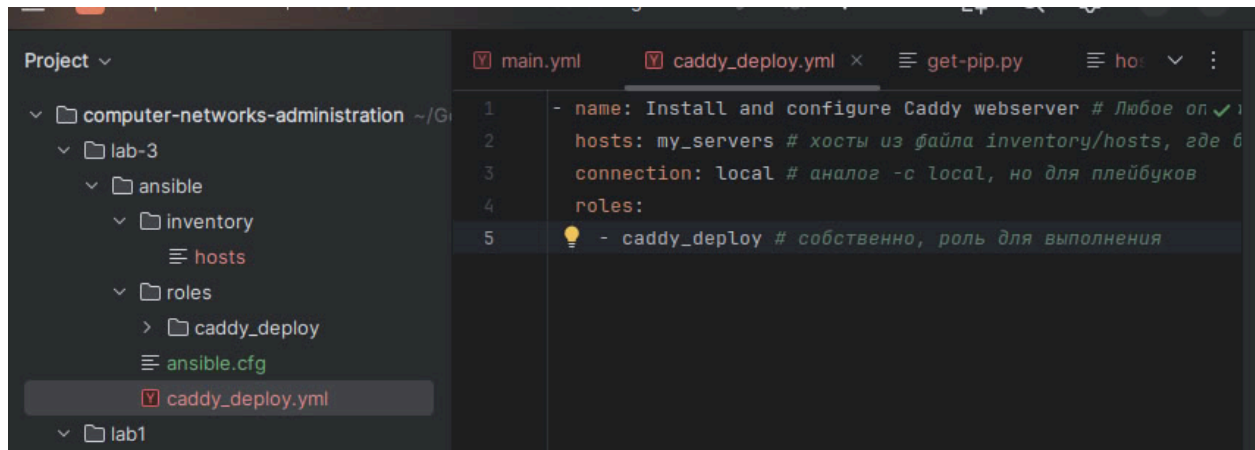
Посмотрим, что получилось

```
ansible-galaxy init caddy_deploy
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles$
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles$
cd caddy_deploy/
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles/
caddy_deploy$ ls
defaults  files  handlers  meta  README.md  tasks  templates  tests  vars
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible/roles/
caddy_deploy$
```

Заполним файл roles/caddy_deploy/tasks/main.yml – конфиг для последовательных задачек для выполнения в playbook



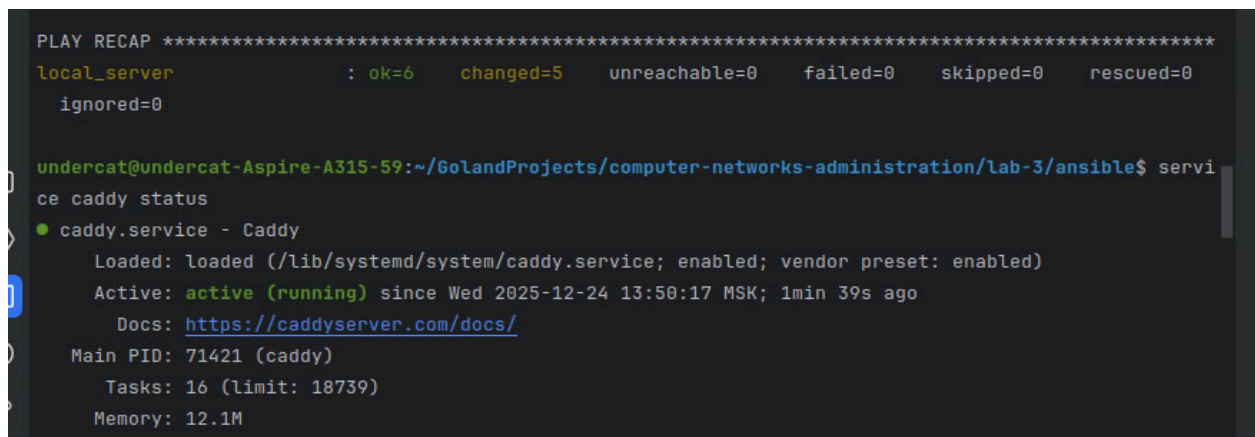
Заполним файл самого playbook-a



The screenshot shows a code editor with a sidebar on the left displaying a project structure. The main editor area shows the content of the file `caddy_deploy.yml`. The file contains a single play with the following details:

- `name`: Install and configure Caddy webserver # Любое оп
- `hosts`: my_servers # хосты из файла inventory/hosts, где б
- `connection`: local # аналог -с local, но для плейбуков
- `roles`:
 - `caddy_deploy` # собственно, роль для выполнения

Запустим плейбук, проверим статус



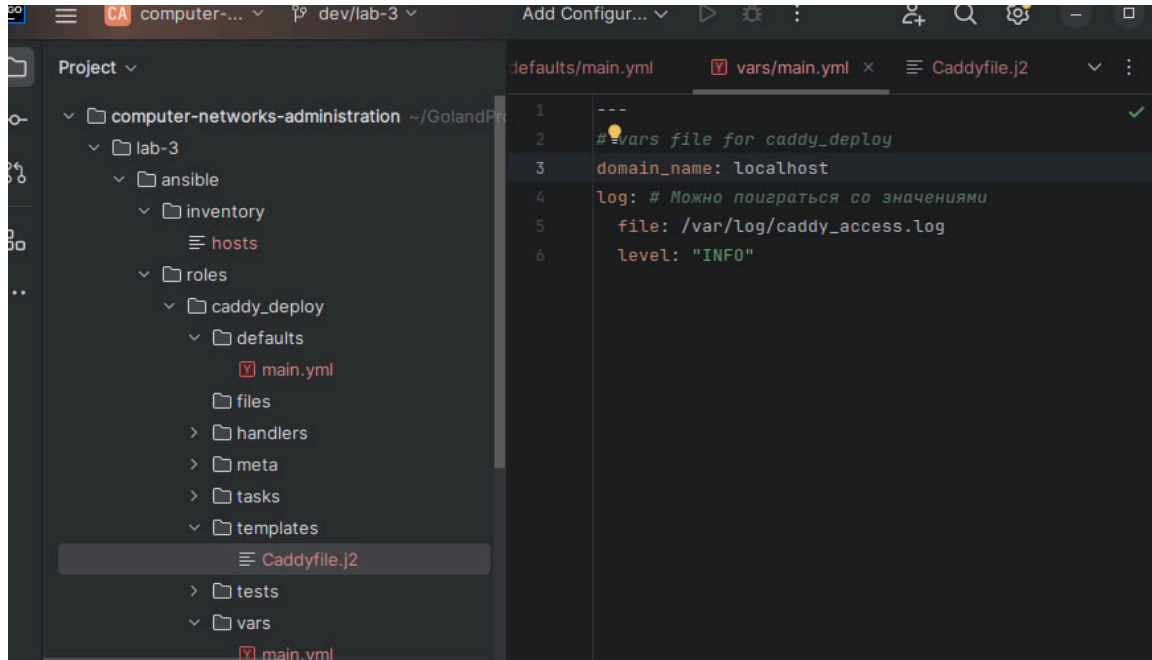
The screenshot shows a terminal window with the following output:

```
PLAY RECAP *****
local_server      : ok=6   changed=5   unreachable=0   failed=0   skipped=0   rescued=0   ignored=0

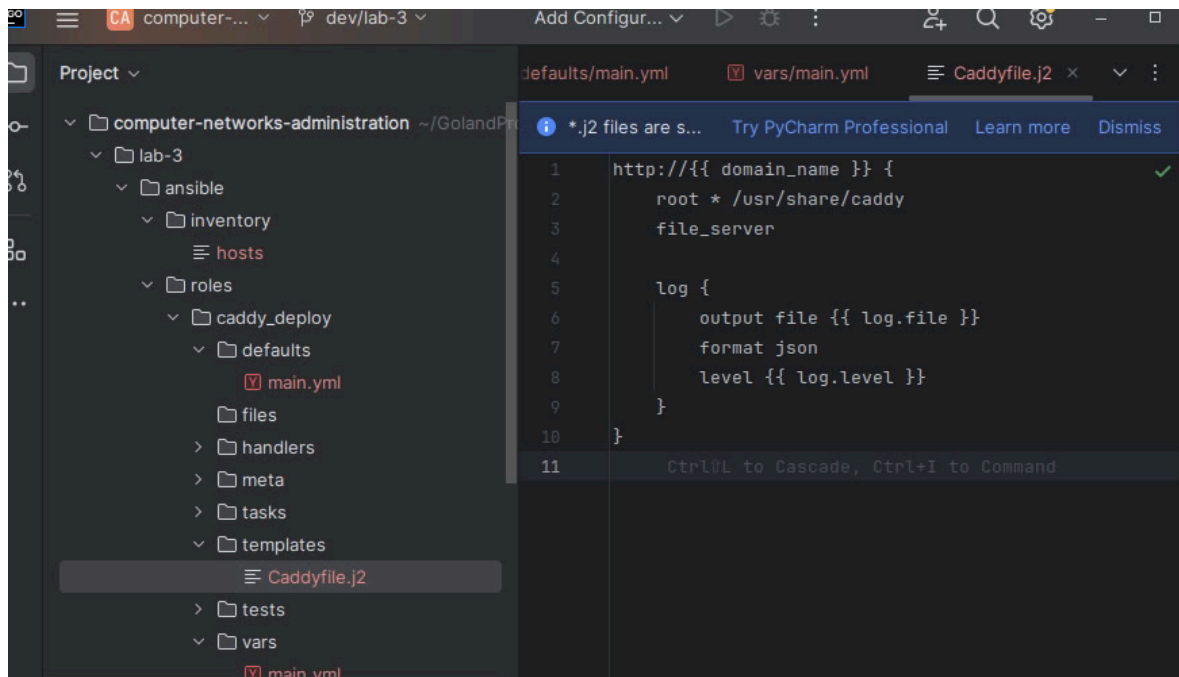
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ service caddy status
● caddy.service - Caddy
   Loaded: loaded (/lib/systemd/system/caddy.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2025-12-24 13:50:17 MSK; 1min 39s ago
     Docs: https://caddyserver.com/docs/
   Main PID: 71421 (caddy)
    Tasks: 16 (limit: 18739)
   Memory: 12.1M
```

Часть 3 (запускаемся локально)

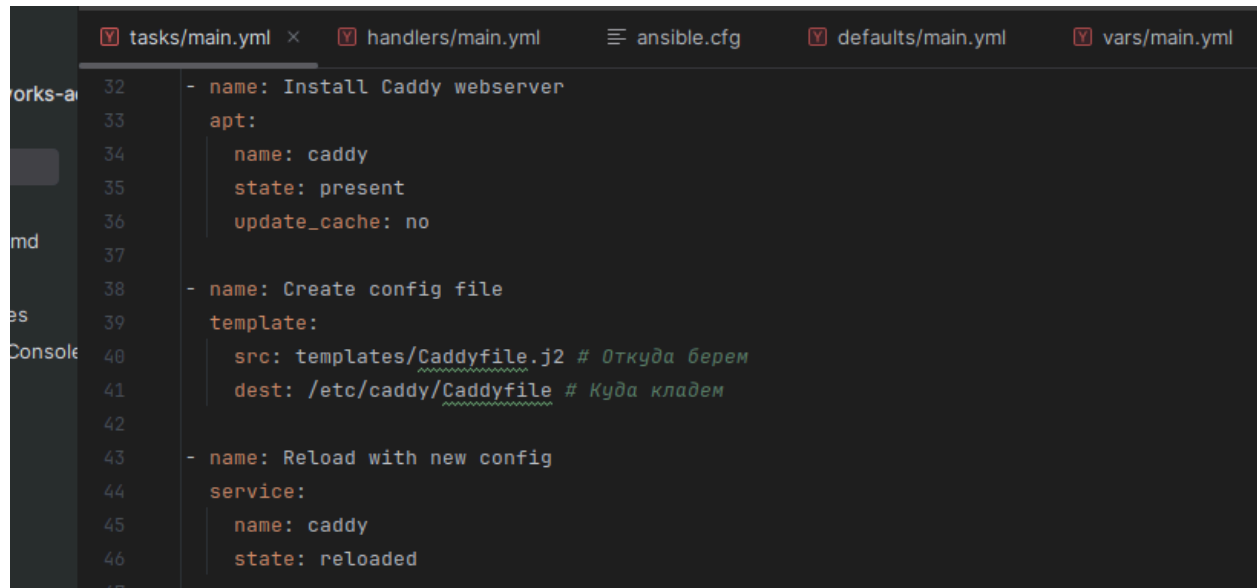
Укажем, куда складывать логи и на каком домене будет доступен caddy.



Сделаем соотв. шаблон .j2, по которому будет создан далее реальный Caddyfile



Чтобы файл генерировался на основании шаблона и переменных, добавим в tasks новую задачу на создание файла и перезагрузку конфигурации caddy

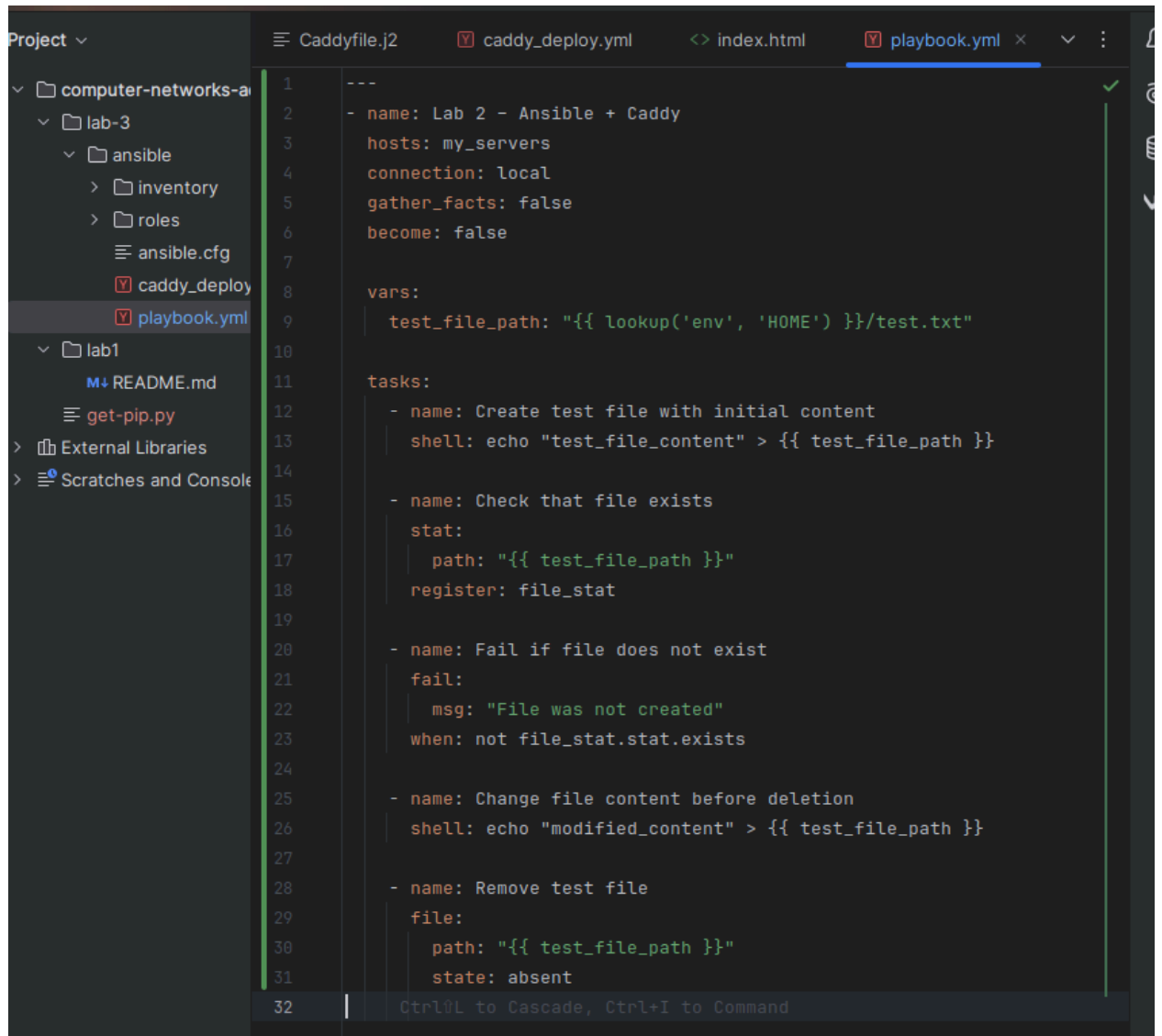


```
tasks/main.yml x handlers/main.yml ansible.cfg defaults/main.yml vars/main.yml
32 - name: Install Caddy webserver
33   apt:
34     name: caddy
35     state: present
36     update_cache: no
37
38 - name: Create config file
39   template:
40     src: templates/Caddyfile.j2 # Откуда берем
41     dest: /etc/caddy/Caddyfile # Куда кладем
42
43 - name: Reload with new config
44   service:
45     name: caddy
46     state: reloaded
47
```

Перезагрузим caddy, посмотрим на стартовый экран.

Задание

- 1) Создадим новый playbook со следующими задачами: создать файл, проверить файл, фейлануться если файл не нашелся, что-то в нем поменять, удалить



The screenshot shows a code editor with a project sidebar on the left and a main editor area. The sidebar shows a project structure with folders like 'computer-networks-a', 'lab-3', 'ansible', 'inventory', 'roles', 'ansible.cfg', 'caddy_deploy', 'playbook.yml', 'lab1', 'README.md', 'get-pip.py', 'External Libraries', and 'Scratches and Console'. The main editor area shows the content of 'playbook.yml' with line numbers from 1 to 32. The playbook is titled 'Lab 2 - Ansible + Caddy' and is configured for 'my_servers' with a local connection. It defines a variable 'test_file_path' and a series of tasks: creating a test file, checking its existence, failing if it doesn't exist, changing its content, and finally removing it.

```
1 ---
2 - name: Lab 2 - Ansible + Caddy
3   hosts: my_servers
4   connection: local
5   gather_facts: false
6   become: false
7
8   vars:
9     test_file_path: "{{ lookup('env', 'HOME') }}/test.txt"
10
11   tasks:
12     - name: Create test file with initial content
13       shell: echo "test_file_content" > {{ test_file_path }}
14
15     - name: Check that file exists
16       stat:
17         path: "{{ test_file_path }}"
18       register: file_stat
19
20     - name: Fail if file does not exist
21       fail:
22         msg: "File was not created"
23       when: not file_stat.stat.exists
24
25     - name: Change file content before deletion
26       shell: echo "modified_content" > {{ test_file_path }}
27
28     - name: Remove test file
29       file:
30         path: "{{ test_file_path }}"
31         state: absent
32
```


Проверим, как оно отработало.

```
undercat@undercat-Aspire-A315-59:~/GoLandProjects/computer-networks-administration/lab-3/ansible$ ansible-playbook playbook.yml

PLAY [Lab 2 - Ansible + Caddy] *****

TASK [Create test file with initial content] *****
[WARNING]: Platform linux on host local_server is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-core/2.17/reference\_appendices/interpreter\_discovery.html for more information.
changed: [local_server]

TASK [Check that file exists] *****
ok: [local_server]

TASK [Fail if file does not exist] *****
skipping: [local_server]

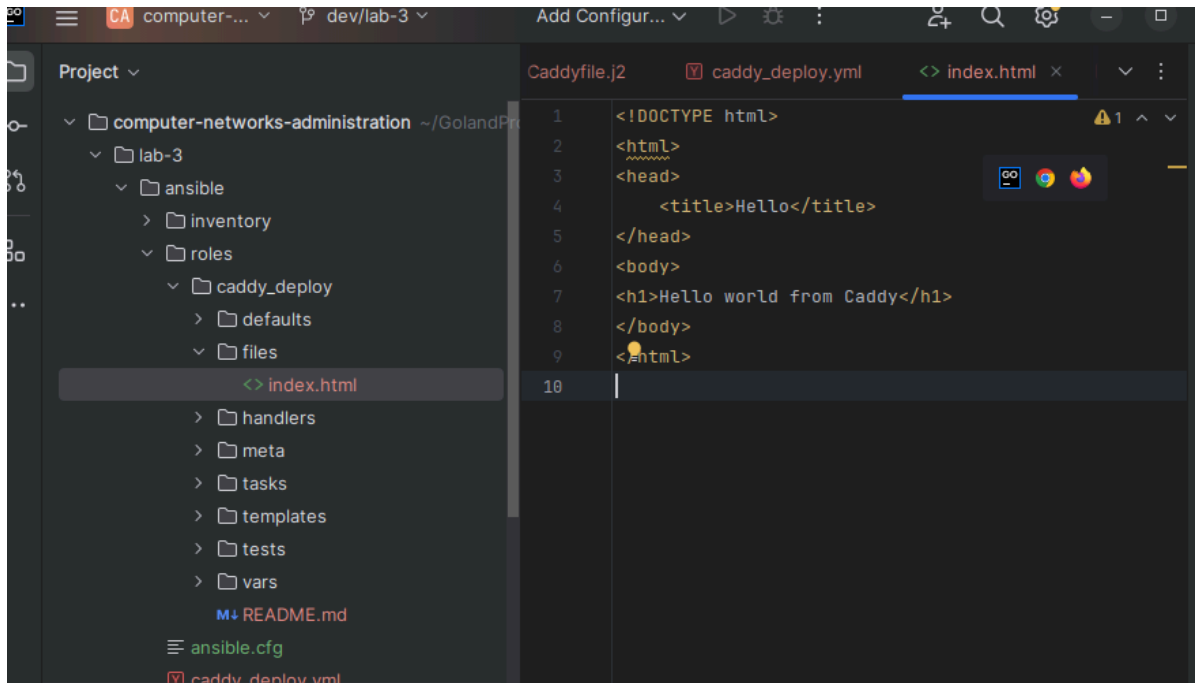
TASK [Change file content before deletion] *****
changed: [local_server]

TASK [Remove test file] *****
changed: [local_server]

PLAY RECAP *****
local_server      : ok=4    changed=3    unreachable=0    failed=0    skipped=1    rescued=0
ignored=0

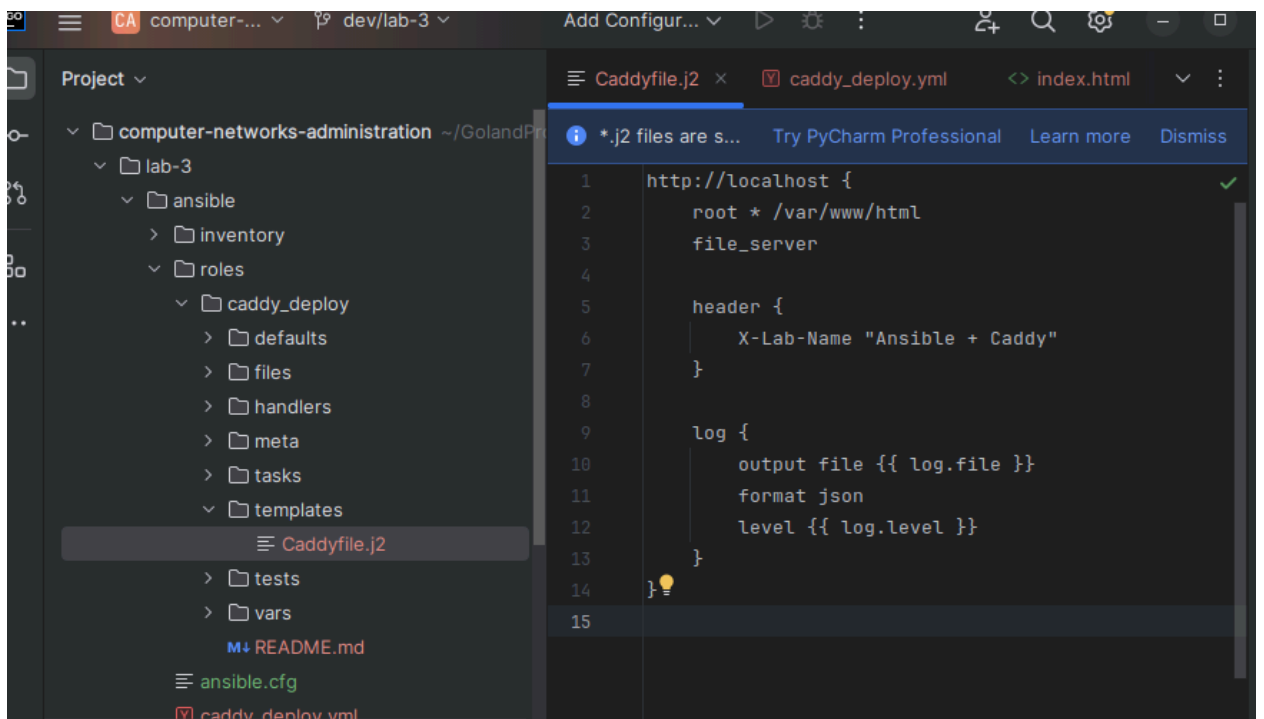
computer-networks-administration > lab-3 > ansible > [V] pla  Windsurf 32:1 LF UTF-8 2 spaces No JSON schema
```

2) Добавим немного логики: добавим стартовую html-страничку



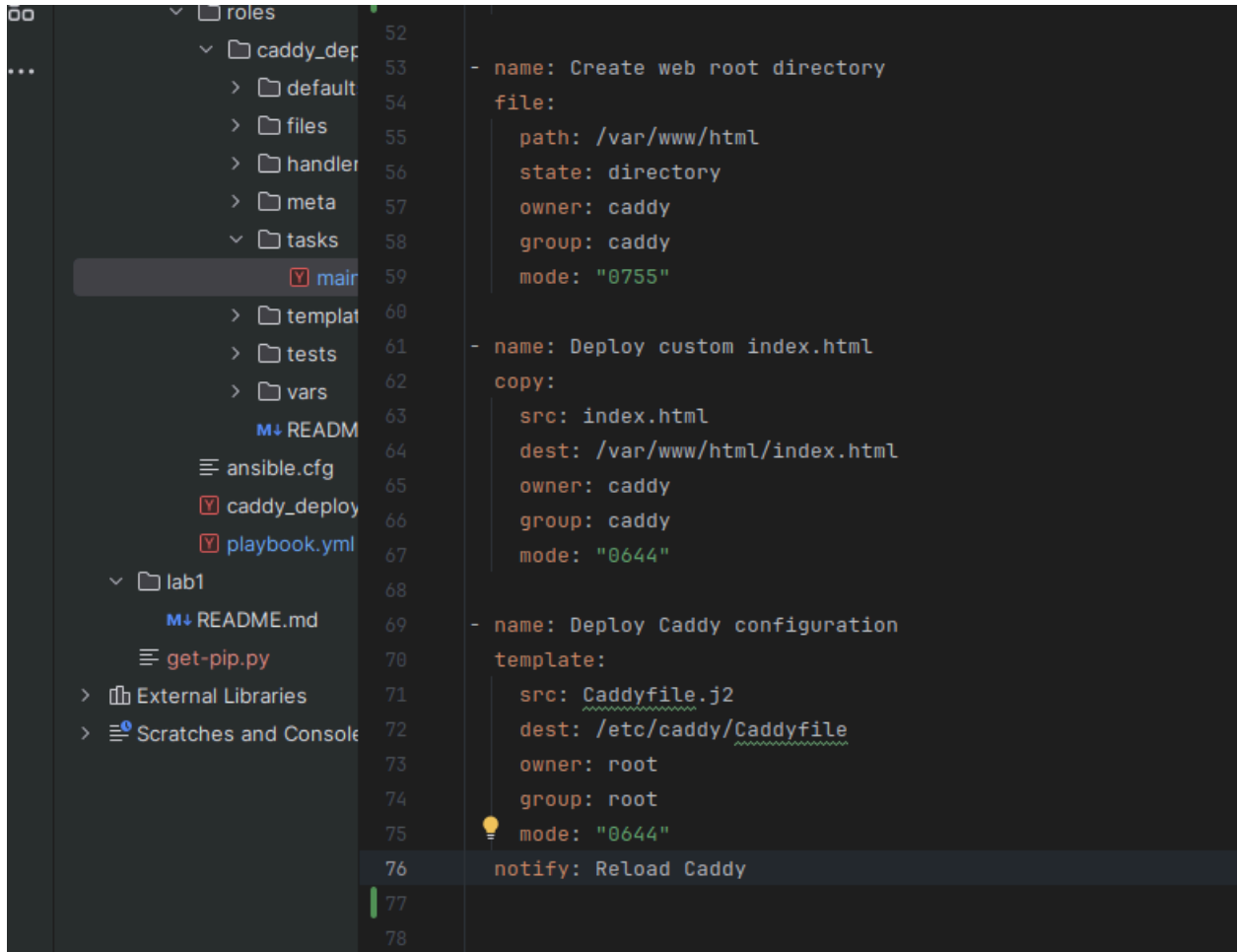
```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Hello</title>
5 </head>
6 <body>
7   <h1>Hello world from Caddy</h1>
8 </body>
9 </html>
```

Добавим кастомный заголовок



```
1 http://localhost {
2   root * /var/www/html
3   file_server
4
5   header {
6     X-Lab-Name "Ansible + Caddy"
7   }
8
9   log {
10    output file {{ log.file }}
11    format json
12    level {{ log.level }}
13  }
14 }
15
```

Добавим новые задачи в исходный playbook: создадим корневую папку сайта, откуда caddy и подцепит html-страничку, копирование в неё самой странички, а также повторный деплой конфигурации caddy



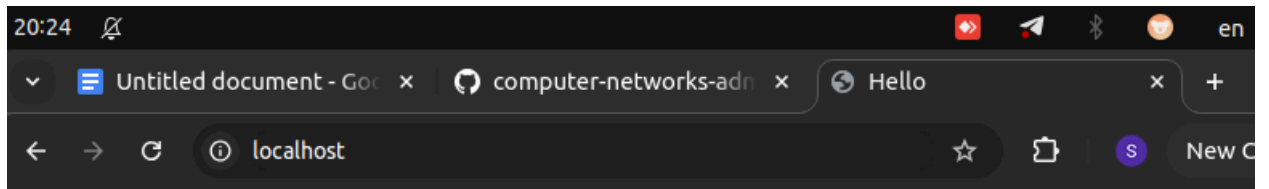
```
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- name: Create web root directory
  file:
    path: /var/www/html
    state: directory
    owner: caddy
    group: caddy
    mode: "0755"

- name: Deploy custom index.html
  copy:
    src: index.html
    dest: /var/www/html/index.html
    owner: caddy
    group: caddy
    mode: "0644"

- name: Deploy Caddy configuration
  template:
    src: Caddyfile.j2
    dest: /etc/caddy/Caddyfile
    owner: root
    group: root
    mode: "0644"
  notify: Reload Caddy
```

Перезапустим, проверим :) :) :)



Hello world from Caddy

You will be redirected shortly...

```
undercat@undercat-Aspire-A315-59:~/Downloads/GoLand-2024.3/bin$ curl -I http://localhost
HTTP/1.1 200 OK
Accept-Ranges: bytes
Content-Length: 356
Content-Type: text/html; charset=utf-8
Etag: "df6lcofdj4xm9w"
Last-Modified: Wed, 24 Dec 2025 16:35:23 GMT
Server: Caddy
Vary: Accept-Encoding
X-Lab-Name: Ansible + Caddy
Date: Wed, 24 Dec 2025 17:26:34 GMT
```